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| **Branch:** MCA (Data Science) Kargil | **Semester:** 2 |
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# Experiment 10

**Experiment Aim:**

To implement Transaction Control Language (TCL) commands in PostgreSQL using COMMIT, ROLLBACK, and SAVEPOINT for maintaining data consistency and controlling transactions.

**Tools used:** PostgreSQL

**Objectives:**

- **Transaction Management:** To understand the use of TCL commands in PostgreSQL.
- **Rollback Operations:** To revert unwanted changes using ROLLBACK.
- **Savepoint Usage:** To create checkpoints within transactions using SAVEPOINT.
- **Data Integrity:** To maintain consistency during multiple database operations.

**Theory:**

Transaction Control Language (TCL) commands are used to manage transactions in a database. A transaction is a sequence of SQL operations performed as a single logical unit. PostgreSQL supports TCL commands such as COMMIT, ROLLBACK, and SAVEPOINT to ensure reliability, consistency, and recovery in case of errors.

**Experiment Steps:**

*Step 1: Creating Table*

**Query:**

```
CREATE TABLE Payroll (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2) CHECK(SALARY > 0)
);
```

*Step 2: Inserting Sample Data*

**Query:**

```
INSERT INTO Payroll VALUES
(1, 'Amit', 30000),
(2, 'Neha', 40000),
(3, 'Ravi', 50000);
```

SELECT * FROM Payroll;

**Output:**

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | emp_id [PK] integer | emp_name character varying (50) | salary numeric (10,2) |
|---|---|---|---|
| 1 | 1 | Amit | 30000.00 |
| 2 | 2 | Neha | 40000.00 |
| 3 | 3 | Ravi | 50000.00 |

*Step 3: Using ROLLBACK*

**Query:**

```
-- using rollback
begin;

UPDATE Payroll
SET salary = -1000
WHERE emp_id = 3;

ROLLBACK;
```

UPDATE Payroll

SET salary = 1000

WHERE emp_id = 3;

SELECT * FROM Payroll

**Output:**

Data Output Messages Notifications

```
ERROR:  new row for relation "payroll" violates check constraint
"payroll_salary_check"
Failing row contains (3, Ravi, -1000.00).

SQL state: 23514
Detail: Failing row contains (3, Ravi, -1000.00).
```

Data Output Messages Notifications

```
ROLLBACK

Query returned successfully in 131 msec.
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | emp_id [PK] integer | emp_name character varying (50) | salary numeric (10,2) |
|---|---|---|---|
| 1 | 1 | Amit | 30000.00 |
| 2 | 2 | Neha | 40000.00 |
| 3 | 3 | Ravi | 1000.00 |

*Step 4: Using SAVEPOINT*

**Query:**

```
BEGIN;

UPDATE Payroll
SET salary = salary + 5000
WHERE emp_id = 1;

SAVEPOINT sp1;

UPDATE Payroll
SET salary = salary + 7000
```

```
WHERE emp_id = 2;

UPDATE Payroll
SET salary = -1000
WHERE emp_id = 3;

ROLLBACK TO sp1;

SELECT * FROM Payroll;

COMMIT;
```

**Output:**

```
Data Output   Messages   Notifications
UPDATE 1

Query returned successfully in 529 msec.
```

```
Data Output   Messages   Notifications
SAVEPOINT

Query returned successfully in 112 msec.
```

Data Output Messages Notifications

```
ERROR:  new row for relation "payroll" violates check constraint
"payroll_salary_check"
Failing row contains (3, Ravi, -1000.00).

SQL state: 23514
Detail: Failing row contains (3, Ravi, -1000.00).
```

Data Output Messages Notifications

```
ROLLBACK

Query returned successfully in 81 msec.
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

| | emp_id [PK] integer | emp_name character varying (50) | salary numeric (10,2) |
|---|---|---|---|
| 1 | 2 | Neha | 40000.00 |
| 2 | 3 | Ravi | 1000.00 |
| 3 | 1 | Amit | 35000.00 |

## Outcomes:

- Students will be able to use TCL commands in PostgreSQL.
- Students will understand transaction rollback mechanisms.
- Students will learn to use SAVEPOINT for partial rollback.
- Students will maintain data consistency during transactions.